

Introduction to Data Processing

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Lecture 2

Handling Data

Data Wrangling

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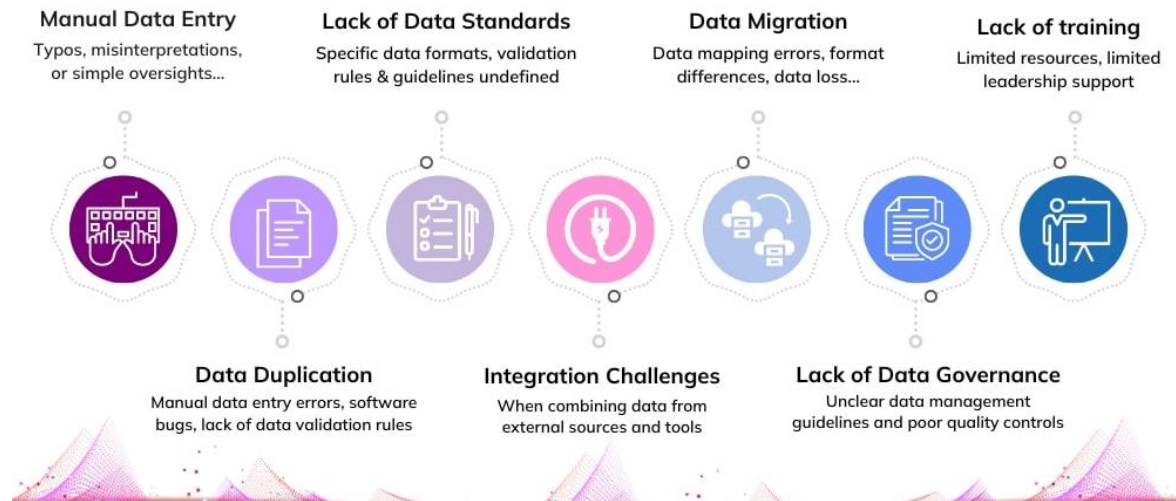
- Loading a dataset for the first time (sanity checks, data parsing issues, ...)
- Manipulating different types of data (text, dates, categorical, images)
- Feature engineering
- Missing data
- Privacy and anonymisation



Data Consistency

- Having a First Look at the Data (open the files and look at the content!)
- Null Values
- Unexpected Column Types
- Sanity Checking Values
 - Numeric Columns
 - Text and Categorical Columns
 - Duplicate Data
 - Displaying Data Frames with Style

Why Data Inconsistency Occur?

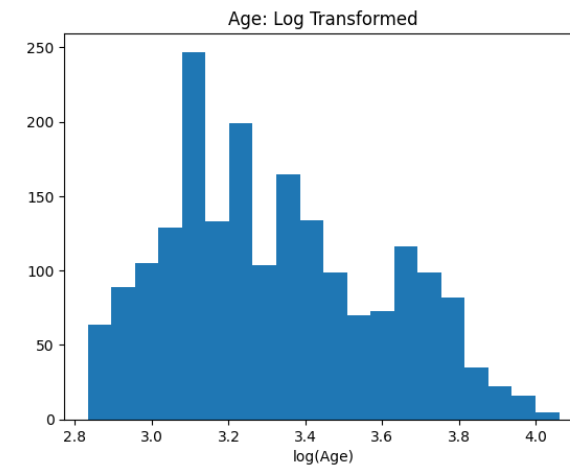
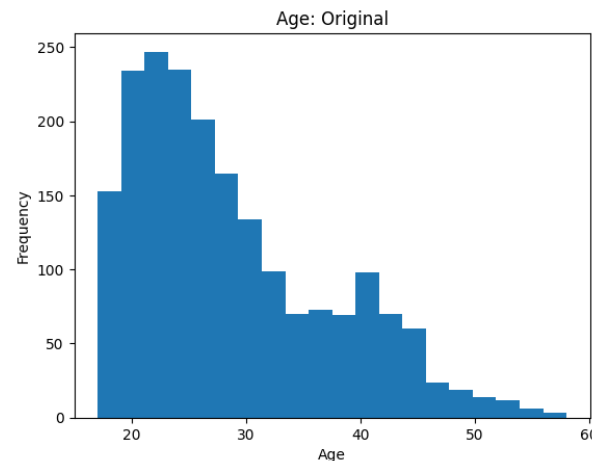


Modifying Columns and Indices

- Dropping Named Columns and Indices
 - We may wish to drop columns or indices entirely from a DataFrame.
- Dropping Columns and Indices with Nulls
 - It may be that we wish to drop columns according to whether they have missing data.
- Relabelling Columns and Indices
- Sometimes dealing with data that is inconveniently named.

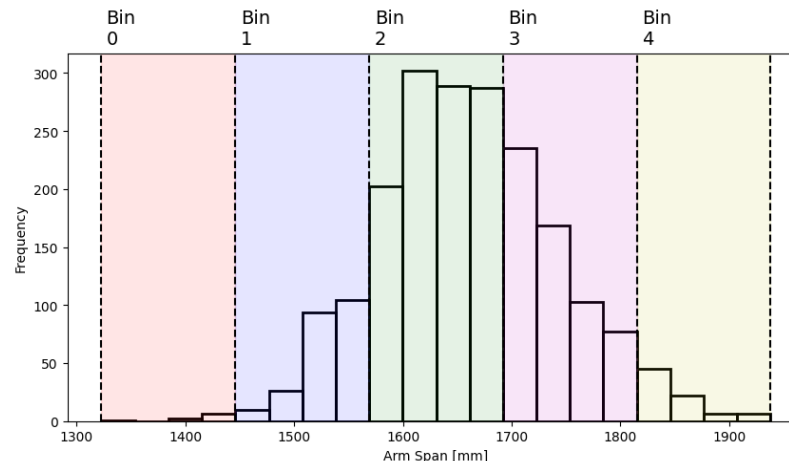
Scaling & Transforming Features

- Converting Units
- Normalisation (imposed min, max or average)
- Standardisation
 - Another popular scaling approach is to “standardise” the data, or transform it so the feature has a mean of zero and a standard deviation of one
 - Achieved by subtracting the original mean and dividing by the original standard deviation of the feature
- Transformation



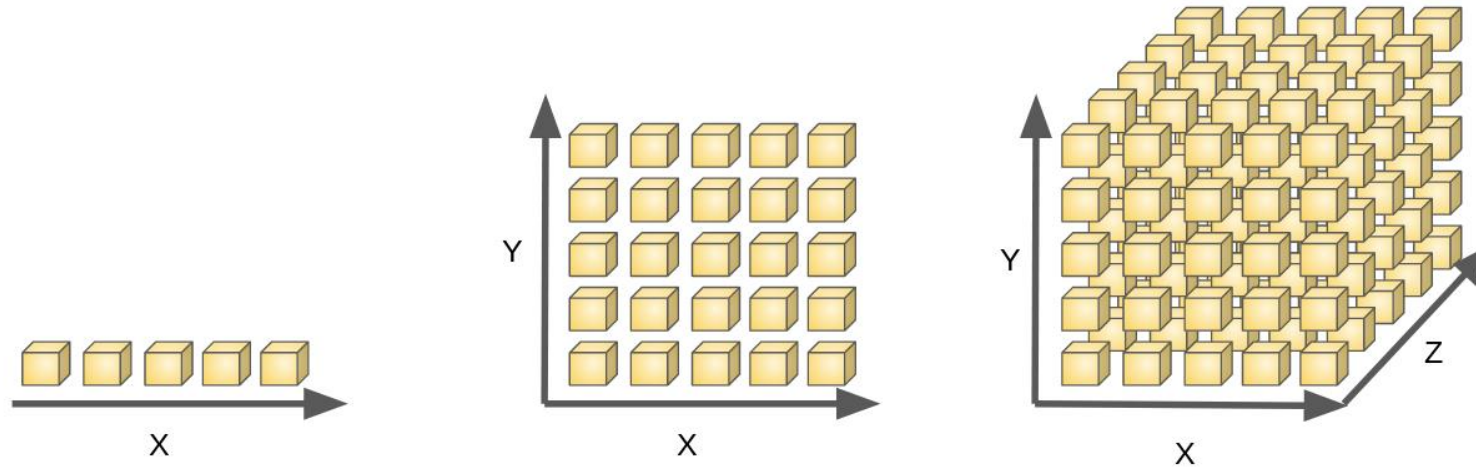
Feature Engineering

- Creating New Features (**more details in lecture 4**)
 - Another common task is to use domain knowledge to create new features that may be of interest for an analysis, or could add predictive power to a model.
- Binning
 - Binning is the process of converting a column of continuous values into a discrete set of categories



Feature Selection and Dimensionality Reduction

- Another important topic is the “curse of dimensionality”, or what to do when we have many features (columns) relative to the number of samples (rows) in the dataset.



- Assume that we only need 5 data points to uniformly sample the data in 1 dimension.
- However, to maintain equal distance between points in both dimensions, it's required 25 points.
- And for 3 dimensions, we need $5^3 = 125 \approx 10^2$ points.
- So, if we have 10 dimensions, we need $5^{10} \approx 10^7$ data points to achieve uniform sampling

Feature Selection and Dimensionality Reduction

- Broadly speaking there are three techniques that can be used when this is a problem:
 - **Feature selection:** Only include a subset of the available features in the model, keeping those that are the strongest predictors for the target variable.
 - Scikit-learn has some algorithms for feature selection available.
 - **Dimensionality reduction:** Condense the original features into a smaller number of new features (each one being a function/combination of the original features) that maintain important relationships in the data.
 - The most well-known technique here is Principal Component Analysis, which also has a scikit-learn implementation.
 - **Regularisation:** Modifies the “cost function” of a model to penalise complexity (i.e., preferring solutions that give a large weighting to a smaller number of features).

Data Manipulation

- Time and Date
- Text
- Categorical
- Image

Time and Date Data

- Dealing with time and date data can be tricky.
- String-formatted data is hard to compare and represent for modelling.

```
print("02/03/1900" > "01/01/2020")  
True
```

- We need to represent date data in a format that will allow us to compare items and perform operations such as addition and subtraction.
- Python's standard library includes the datetime module. This allows us to represent dates and times as structured objects.

```
import datetime  
# create a datetime object with value set to now  
now = datetime.datetime.now()  
print(now)
```

Text data

- Inconsistencies
 - Strings have their own particular array of consistency issues, such as inconsistent capitalisation and extraneous whitespace.
- Spelling is tricky
 - In a relatively simple scenario, with categorical data encoded as strings, you might be able to spot these by checking for all unique values in your data.
- Splitting
 - We'll also commonly want to split a string based on a particular delimiter or separator. For example, we may wish to split a string of text into individual words, using any whitespace separator.

Text data

- Joining

```
my_list = ["a", "list", "of", "words"] # join with a space separator  
s = " ".join(my_list)
```

- Regular expressions

- Regular expressions (regexps, regex) are character sequences that specify a search pattern, usually for a find and/or replace task on text data.

```
import re  
txt = "The rain in Spain falls mainly on the plains. So they say, anyway."  
# find all words starting with upper case S or lower case t  
print(re.findall(r"\b[St]\w+", txt))
```

- NLP Preprocessing

- In Natural Language Processing (NLP) tasks we often see some slightly more complicated preprocessing such as:
 - Stemming and Lemmatisation - reducing words to common base forms
 - Stop-word Removal - removing common words that carry little information
 - “Vectorization” - convert text to a meaningful numeric vector representation (e.g. term frequency encoding)

Categorical Data

- Categorical variables relate to data that is summarised in groups or has a limited set of possible values.
- Examples could be t-shirt size (small, medium, large, ...), degree grade (first class, second class, ...), or car manufacturer (Ford, Toyota, ...).

	Program Category	Region Served	Age Group	Sex	Living Situation	Household Composition	Preferred Language	Veteran Status
0	INPATIENT	HUDSON RIVER REGION	ADULT	FEMALE	INSTITUTIONAL SETTING	NaN	ENGLISH	NO
1	SUPPORT	WESTERN REGION	CHILD	MALE	PRIVATE RESIDENCE	COHABITATES WITH OTHERS	ENGLISH	NO
2	OUTPATIENT	WESTERN REGION	CHILD	FEMALE	PRIVATE RESIDENCE	COHABITATES WITH OTHERS	ENGLISH	NO
3	OUTPATIENT	NEW YORK CITY REGION	CHILD	FEMALE	PRIVATE RESIDENCE	COHABITATES WITH OTHERS	ENGLISH	NO
4	OUTPATIENT	LONG ISLAND REGION	CHILD	FEMALE	PRIVATE RESIDENCE	COHABITATES WITH OTHERS	ENGLISH	NO

5 rows x 32 columns

Numerical Encodings for Categorical Data

- Binary Data
- Data with Multiple Categories
 - Ordinal Data
 - One-Hot Encoding
 - Grouping Categories

Binary Data

- `df["Smokes"].value_counts()`
NO 3387
YES 1387
Name: Smokes, dtype: int64
- `yesno_encoding = {`
 "YES": 1,
 "NO": 0,
 }
• `df["Smokes"] = df["Smokes"].replace(yesno_encoding)`
- `df["Smokes"].value_counts()`
0.0 3387
1.0 1387
Name: Smokes, dtype: int64

Ordinal Data

- `df["Education Status"].value_counts()`
MIDDLE SCHOOL TO HIGH SCHOOL 2585
SOME COLLEGE 737
COLLEGE OR GRADUATE DEGREE 699
PRE-K TO FIFTH GRADE 433
OTHER 68
NO FORMAL EDUCATION 24
Name: Education Status, dtype: int64
- `edu_encoding = {`
"NO FORMAL EDUCATION": 0,
"PRE-K TO FIFTH GRADE": 1,
"MIDDLE SCHOOL TO HIGH SCHOOL": 2,
"SOME COLLEGE": 3,
"COLLEGE OR GRADUATE DEGREE": 4,
"OTHER": np.nan,
}
• `df["Education Status"] = df["Education Status"].replace(edu_encoding)`

One-Hot Encoding

- `df["Region Served"].value_counts()`
NEW YORK CITY REGION 2486
WESTERN REGION 847
HUDSON RIVER REGION 775
CENTRAL NY REGION 534
LONG ISLAND REGION 358
Name: Region Served, dtype: int64

- If we assigned each region a number from 1-5, an algorithm could try to infer relationships that don't exist.
- For example, we may encode WESTERN_REGION = 2 and CENTRAL NY REGION = 4, but this means we're representing Central NY as "double" Western NY - double what?
- Instead, a common choice is to "one-hot" encode the column, converting it into a binary column for each category.

One-Hot Encoding

- `pd.get_dummies(df["Region Served"])`

	CENTRAL NY REGION	HUDSON RIVER REGION	LONG ISLAND REGION	NEW YORK CITY REGION	WESTERN REGION
0	0	1	0	0	0
1	0	0	0	0	1
2	0	0	0	0	1
3	0	0	0	1	0
4	0	0	1	0	0
...	...	...	...	...	...
4995	0	0	0	0	1
4996	0	0	0	0	1
4997	1	0	0	0	0
4998	1	0	0	0	0
4999	0	0	1	0	0

5000 rows x 5 columns

Grouping Categories

- `df["Preferred Language"].value_counts()`

ENGLISH	4459
SPANISH	359
INDO-EUROPEAN	55
ASIAN AND PACIFIC ISLAND	27
ALL OTHER LANGUAGES	26
AFRO-ASIATIC	14

Name: Preferred Language, dtype: int64

- If we deal with a question related to finding differences between the care given to English and non-English speakers, we could encode the data as a single binary column
- `prefer_english = (df["Preferred Language"] == "ENGLISH").astype(int)`
- `prefer_english.value_counts()`

1	4459
0	541

Name: Preferred Language, dtype: int64

Which Encoding to Choose?

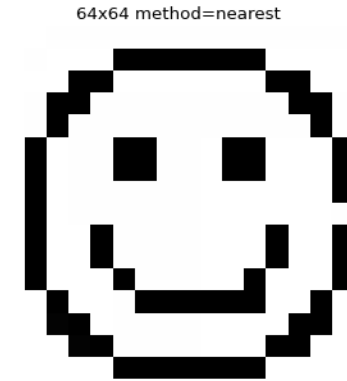
- The encodings described above are the most common, but there are others and there's also no fixed rule that you must use a certain type of encoding for a certain type of data.
- Some implementations
 - Pandas has a categorical data type.
 - The `LabelEncoder`, `OrdinalEncoder` and `OneHotEncoder` classes in the scikit-learn library.

Unseen Values

- If you're developing a model with categorical data, it's important to consider what to do, or how the model will behave, with data from new groups that are not present in the original dataset.
- Say you plan to use a model developed on the 2017 PCS data on new data from 2019, but a new region in New York has been added. How will patients from that region be treated by the model?
- This is essentially a different form of missing data

Image Data

- Normalisation or Standardisation of pixel values
- Resizing (and Resampling)
- Interpolation



Privacy and Anonymisation

- **Anonymisation**

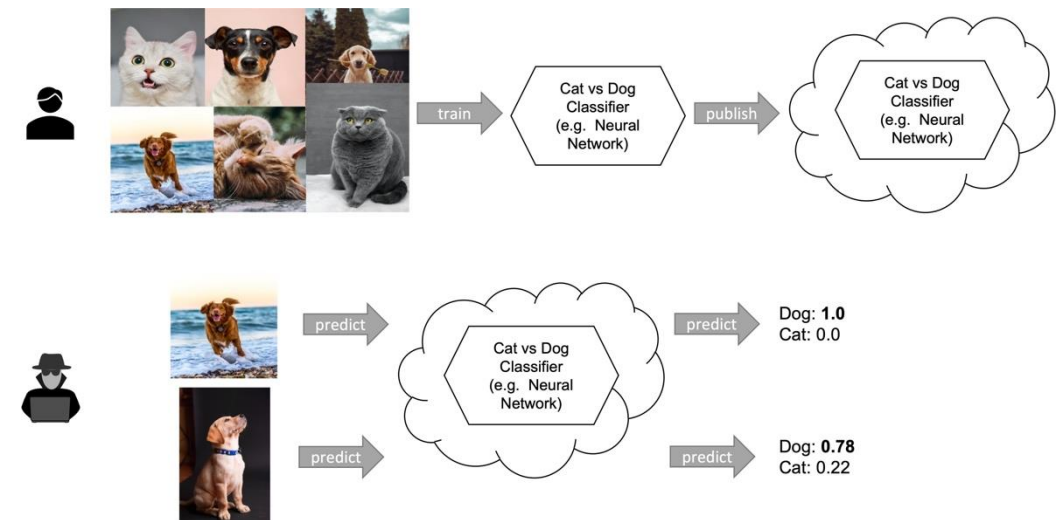
- Anonymisation is the process of removing direct and indirect personal identifiers
- Crucially, post-anonymisation, subjects will not be able to be identified in the data, even given additional information.

- **Pseudonymisation**

- This is subtly different to anonymisation: the subject can be re-identified, given additional information.

- **Differential Privacy**

- first image was part of the original training set
- second image was not part of training training set



Missing Data

- Real world data often has missing values. These may represent gaps in data collection, be expected responses to questions, or be the result of failures or data corruption.

ID	Age	Gender	Name
1	31	F	Alex
2	(?)	M	Ben
3	52	F	Cindy
4	35	(?)	Dan
5	(?)	M	Eric
6	(?)	F	Fay
7	42	M	(?)

- Consider the following question in a survey: “How many days has it been since your flu jab?”
- What answer would you expect from a participant who had not had a flu jab?

Representing Missing Values

- Pandas uses NaN (Not a Number) internally to denote missing data, for reasons of computational speed and convenience (beyond our scope!).
- However, Python's None is also considered a “missing” value and we can check for this with `isna()` or `notna()` methods.

Dealing with Missing Values

- **Listwise Deletion**

- Simply drop any rows that contain missing values. This is the simplest method of dealing with missing data!
- However, this can introduce bias. We should consider why the data is missing. Listwise deletion will introduce bias when the remaining data can no longer be reflective of the original data.

- **Single Imputation**

- Apply a rule to replace a missing value. For example, replace with the mean or the value of the last observation.
- Again, single imputation methods can easily introduce bias. Use with caution.

Missingness Terminology

- **Missing Completely At Random (MCAR)**

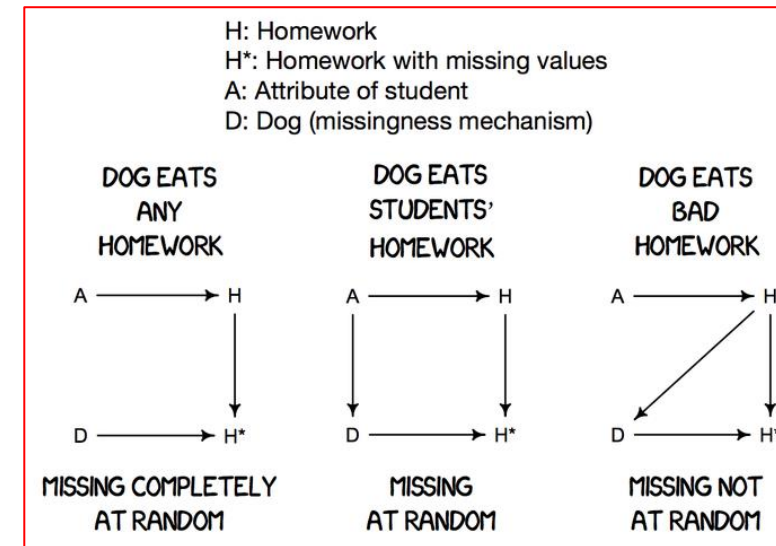
- The missingness is unrelated to both missing and observed values
- Example: Electronic time observations are missing, independent of what lane a swimmer is in.
- If we were to drop rows with MCAR data, we would reduce our population size but would not introduce any bias.

- **Missing At Random (MAR)**

- Missingness is related to observed data but not the missing values themselves
- Example: A certain swimming lane is more likely to have missing electronic time observations
- MAR data allows the prediction of missing values based on complete rows. However, as ever, imputation should be approached with caution!

- **Missing Not At Random (MNAR)**

- The missingness is related to events or factors which are not measured.
- Example : when surveyed people with more income are less likely to report their incomes.
- When data is MNAR no methods exist to handle this missingness appropriately.



Conclusion

Next

- Test several methods
- Compare them
- Keep some of them and clearly justify your choices.